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BSCS 3B

Part A. Identify the ML Problem

Application	Social Media Recommendation
Problem	Recommend contents, accounts, and promotions related to user's preferences
Inputs	Posts liked, search history, followed contents
Output	Recommended contents, posts, products, and influencers
Pattern	Users with similar content, influencers, products, and posts may like to view similar things recommended.
Data	User preferences selected on the social media application/software data.

Part B. Traditional Programming vs. Machine Learning

Traditional Programming

Input-Algorithm-Output

Instructions:

1. When user's search history contains a specific keyword, output posts and accounts tagged with that kind of keyword.
2. If a user follows an account for example, Account A, then output posts from other accounts like Account B and C will be tagged if it is in the same category as Account A.
3. A condition will be made that if a user has liked 5 or more posts under a specific category, then the display algorithm will promote at least 3 or more contents or influencers that are similar or the same to the liked category.
4. Algorithms will prioritize the content based on the previous search history of a user in social media for the last 24 hours.

Machine Learning

Input + Output/Data -> Machine Learning Algorithm

1. Historical user interaction data
2. Watch time the viewers spent in the content

3. Shared posts, sent messages, contents on stories feature
4. Creator Profile Metrics
5. Recent conversations or search terms on other platforms that are linked in your social media accounts \

Part C – Create Your Own Mini Dataset

Input	Output
Watch time percentage of drone videos in a specific social media app is high	Priority score assigned to similar drone contents
Common mutual friends, accounts followed in a common prospective profile	The algorithm will recommend you account that suggested accounts that have the same demographic of followings, mutuals, and contents
Funny videos are commonly shared to a friend	Contents that are the same will be most likely appear on your feed and on your search suggestion
Topics that are detected in the user's profile like photography, adventure, food, aesthetics	Ads or product recommendations frequently appear as suggestions that is based on the data in your profile
Skips specific contents quickly due to preferences	Few to no appearances of that type of contents shown to the user.

Part D

Machine Learning algorithm begins through receiving dataset samples that consist of paired inputs and target outputs. In the duration of training phase, algorithms repeatedly map the inputs given to their corresponding outputs and adjust their internal parameters based on their errors. With the use of error minimization continuously across multiple examples, the algorithm successfully completes learning the relationship or existing pattern within the data. Lastly, trained model uses this learned relationship to make accurate prediction when new and unseen input data was encountered.

Reflection

Learning from data is very useful, instead of manually programming every possible rule is more useful. In this way, we can just look at data trends and patterns and make decisions based on it instead of coding it and prompting the system what to do based on the input of the user. In traditional programming, every block of code matter to the system, every possible rule is necessary, unlike in machine learning, the system learns through data and

predicts output, when we train model, prediction becomes precise overtime as the system becomes exposed to more datasets.